Consortium on Individual Development

samen uniek
TWEELINGONDERZOEK

Road to Open Science

Flux Congress 2022

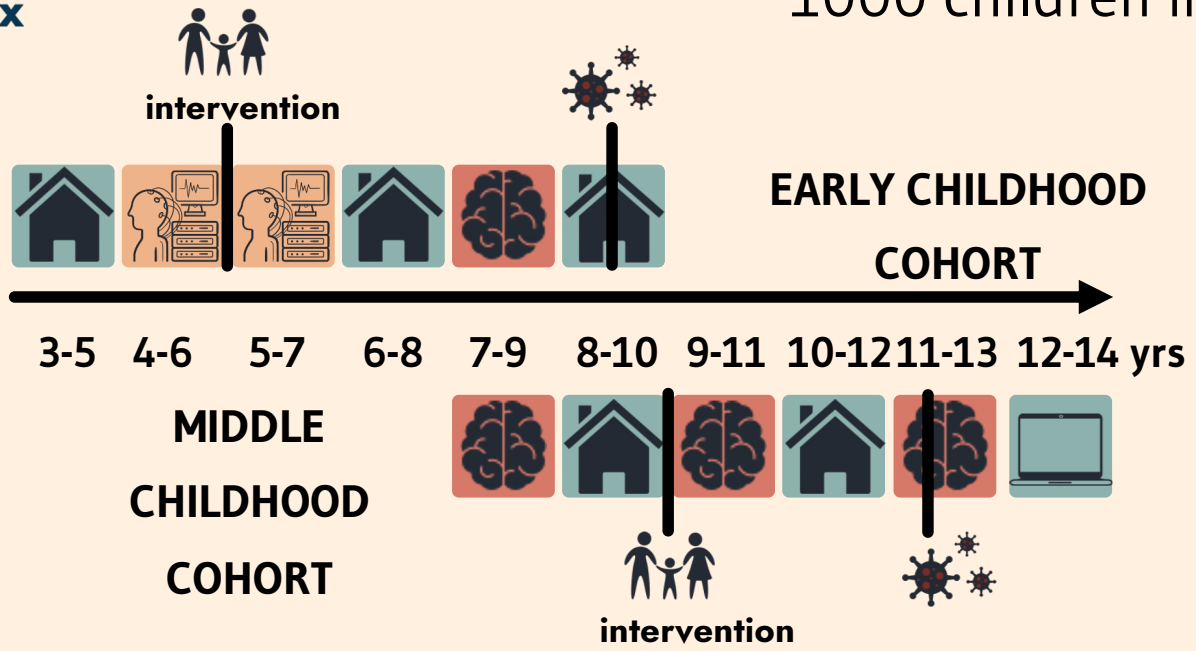
Lara Wierenga, Simone Mulder, & Mark Mulder

www.leidencid.com

L-CID in a nutshell

Multi-informant, multi-method, and multi-index

1000 children in one study



Our team



Prof. Dr.
Eveline Crone



Lara
Wierenga, UD



Michelle
Achterberg, UD



Mara
vd Meulen, PostDoc



Karlijn
Hermans, PostDoc



Stephan
Heunis, PostDoc



Yara
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Simone
Dobbelaar, PhD stud



Lina
van Drunen, PhD stud



Simone
Mulder, Data Manager



Anna
van Duijvenvoorde, UHD



Mark
Mulder, Data Manager

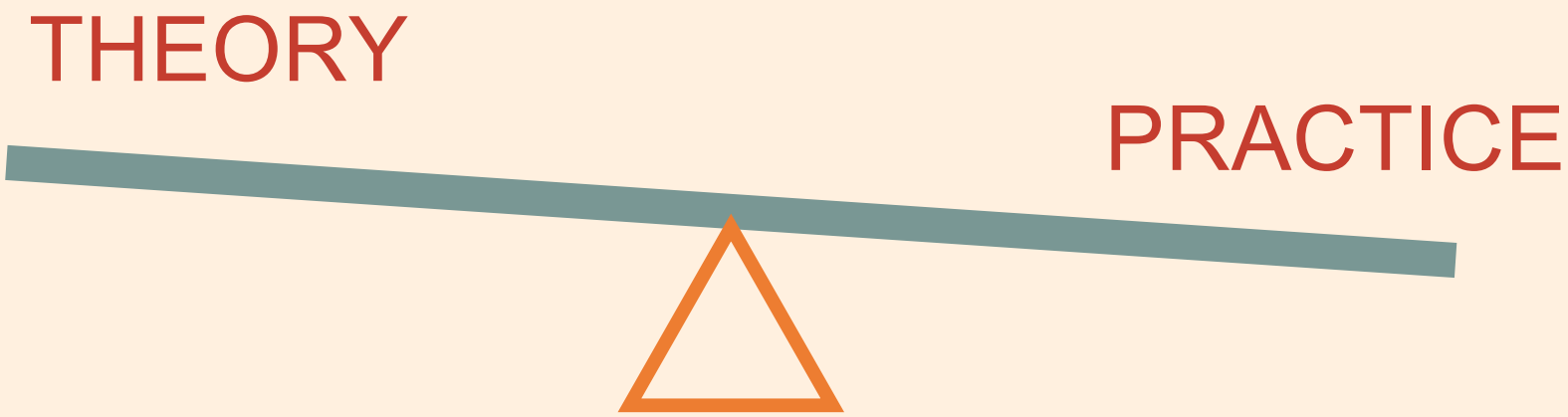
9 Collaborators

35 Alumni

140 Students

Why this workshop?

Open science



Open science vision L-CID

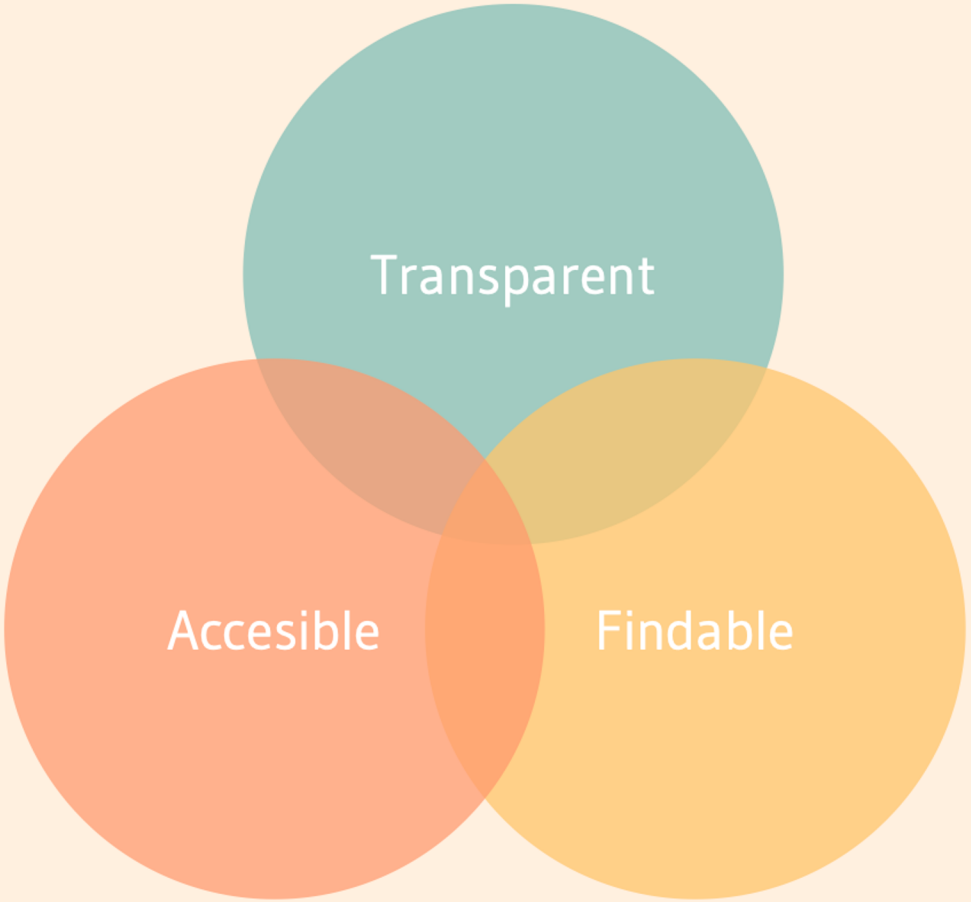
Improve scientific quality

Save limited resources and time

Foster reproducability and replication

Reduce participant burden

Open science vision L-CID



Open science data



www.leidencid.com

1000 measures
1000 participants
>100.000 data files

Citizen Involvement

Outreach (to citizens)

Publish Open access



Open metadata

Open protocols

Open Data

Open code/ open software



Challenge I
How to organize big, multimodal longitudinal data?



Challenge II
How to get from raw data to structured data?



Challenge IV
How to be open and protective?



Challenge V
How to be findable and accessible?



Challenge I: How to organize a big, multimodal longitudinal data set



Our data

- 2 cohorts, 12 waves 
- +- 120 measures
- 3 different types of respondents
- Different types of data 
- Different processing stages
- Metadata files

Total: > 100.000 (!) files to be organized

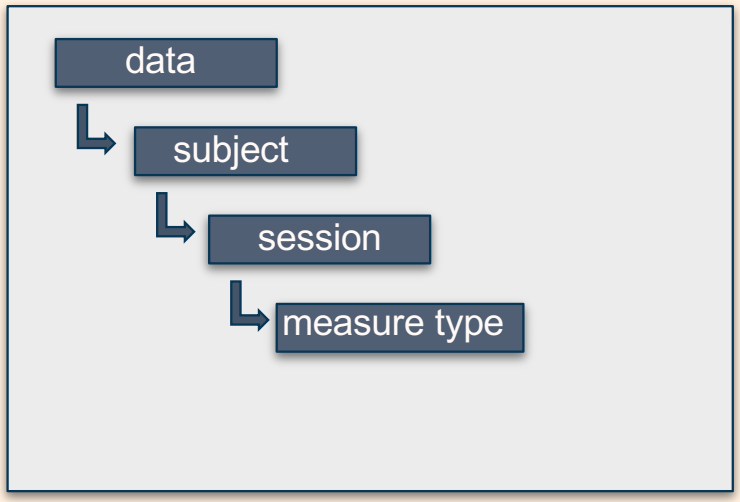
Challenges

1. Standardized format?
2. Data level? Individual vs group level
3. Extensions? -> Should be open access
4. File names, variable names, category names?
5. Processing levels?
6. Metadata?
7. Special cases and exceptions within data?

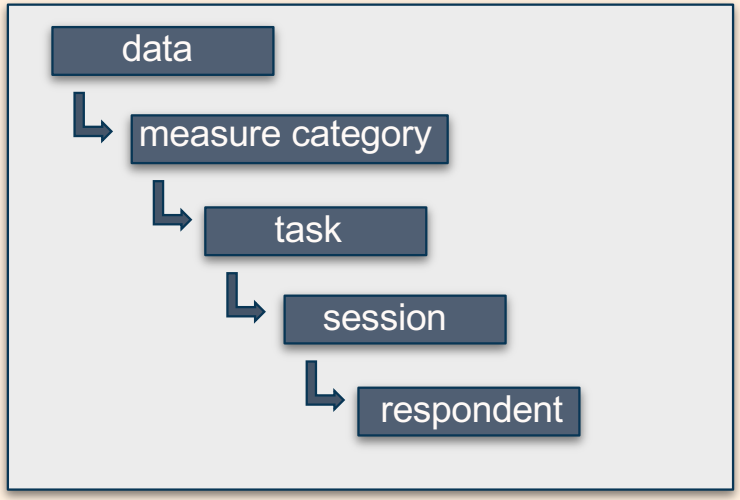
Challenges - Standardized data format

Brain imaging data structure (BIDS) **vs** **Behavioral data structure**

Individual level

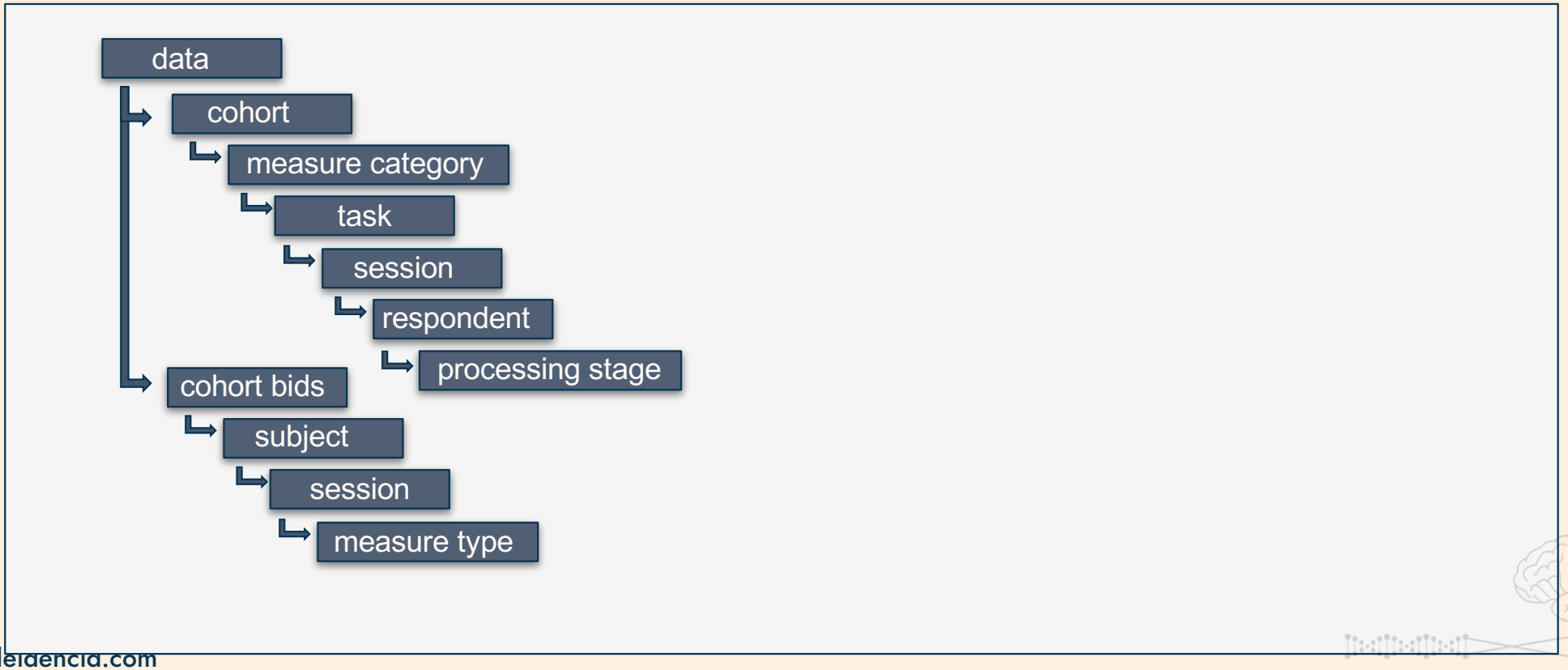


Group level

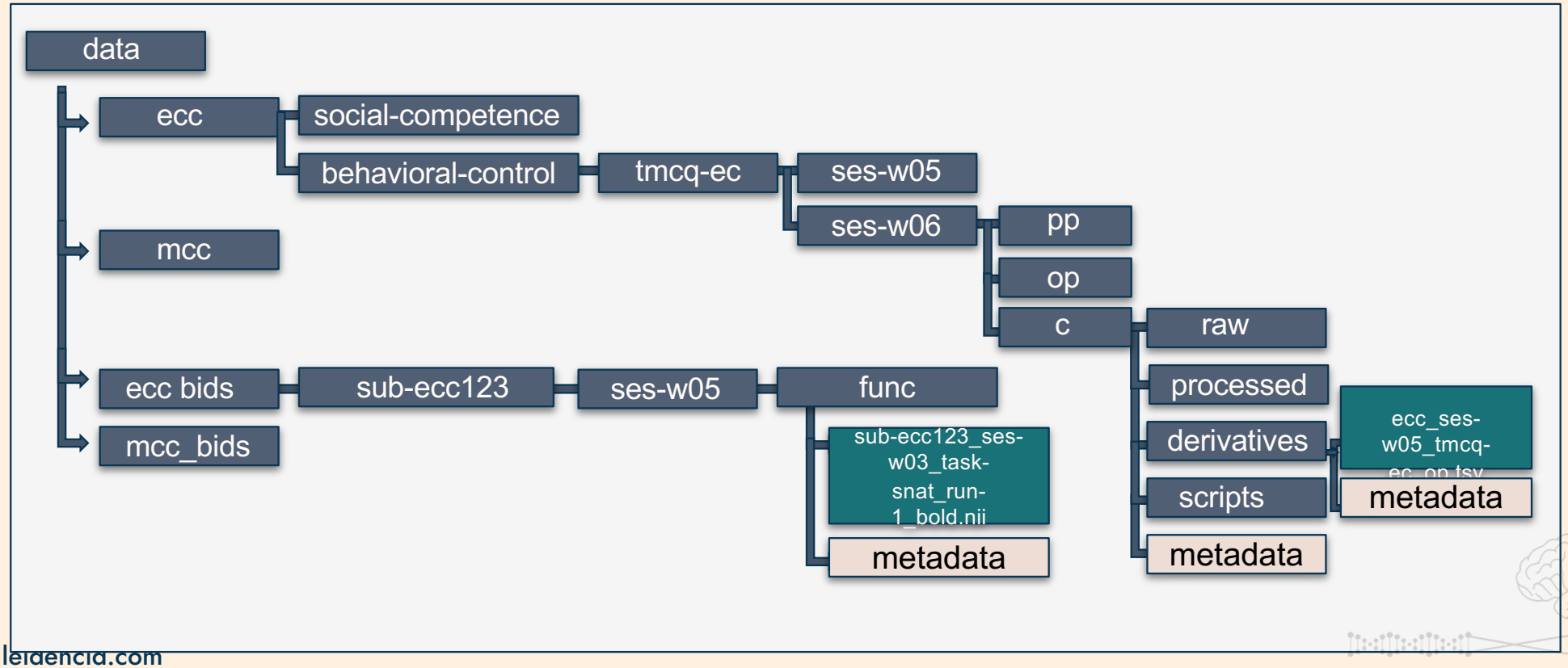


Challenges - Standardized data format

Brain imaging data structure **&** **Behavioral data structure**



Challenges - Structure & naming conventions



Challenges - Metadata



Challenges - Metadata

Info data processing

```
GENERAL INFORMATION

1. Title of derivative data file:  <e.g. ecc_ses-01_sdq_pp.sav/.tsv>   <1 readme file per task & respondent, t

2. Name of JSON file belonging to the datafile:   <e.g. [cohort]_ses-[session]_[task]_[respondent].json>

3. Name(s) of person(s) working on the datafile:

    a. Name:
       Email:

    b. Name:
       Email:
    <etc.>

4. Date of last adjustment to datafile:  <YYYY-MM-DD>

DATA PROCESSING INFORMATION:

1. Timeframe of processing data:
2. General steps of processing data for this file:

    a.                                              <e.g. 1. From raw to processed, the following was done:
                                                          2. From processed to derivatives, ... >
    b.
    c.

     <*If applicable include software used for data processing.>
     <*If applicable include standardized protocols>
     <(briefly explained: can include links to github, scripts, etc.)>

3. Comments: <Special cases/problems/etc.>
```

Challenges – Metadata: JSON files



Possibilities JSON files

1. **General information data file:**

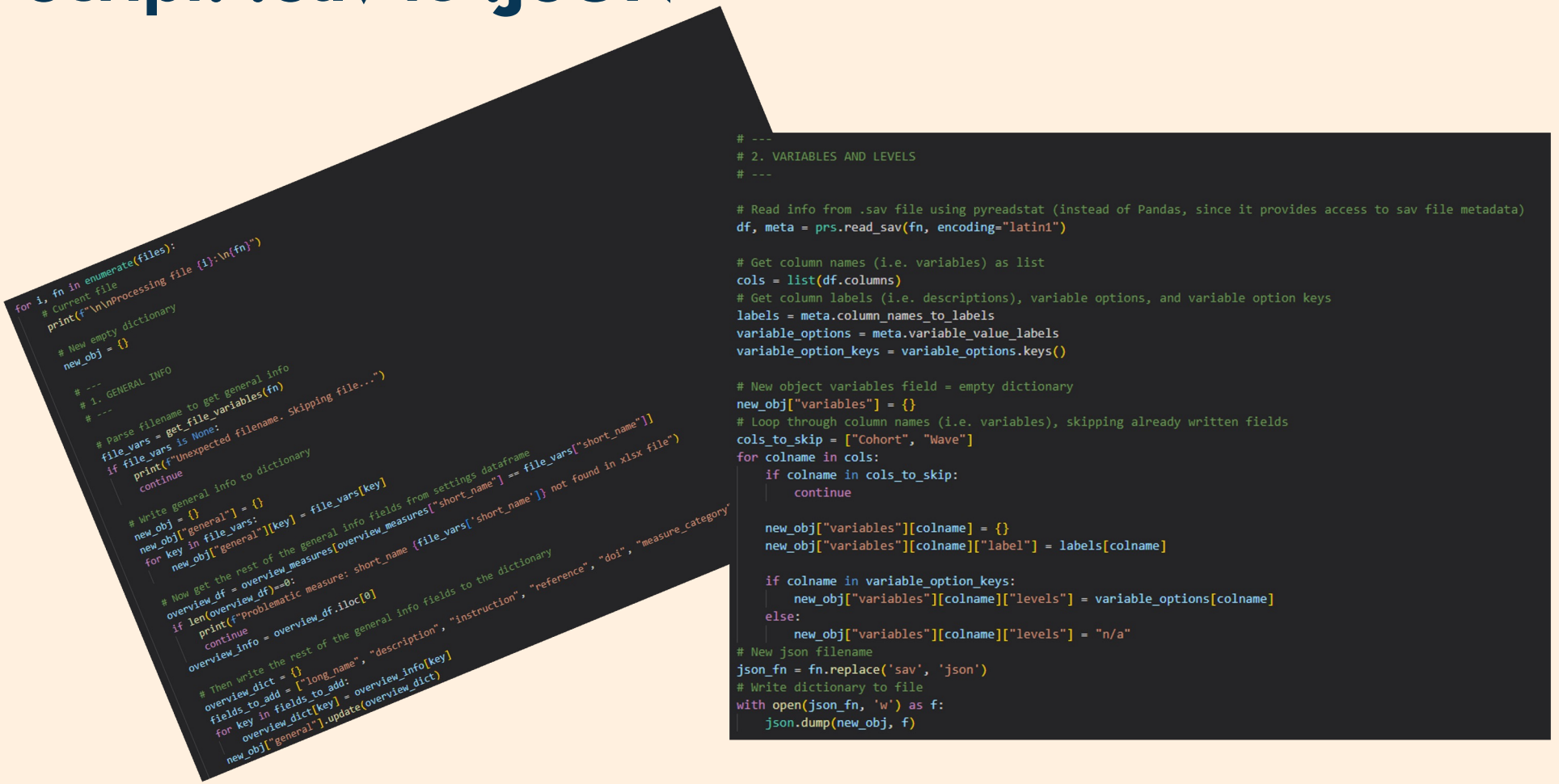
- flexibility and accessibility of data structure
- automating processes: metadata explorer | codebook

Challenges – Metadata: JSON files

Info data itself

```
{
    "general_info": {
        "short_name": "mychild-emp",
        "long_name": "MyChild - Empathy questionnaire.",
        "description": "The MyChild questionnaire is … ",
        "instruction": "..",
        "reference": "..",
        "doi": j.1467-8624.1994.tb00788.x",
        "measure_category": "social-competence",
        "measure_type": "questionnaires",
        "cohort": "ecc" ,
        "wave":   "ses-w05",
        "respondent": "pp"
        },
```

Script: .Sav to .JSON



```
for i, fn in enumerate(files):
    # Current file
    print(f"\n\nProcessing file {i}:\n{fn}")

    # New empty dictionary
    new_obj = {}

    # ---
    # 1. GENERAL INFO
    # ---

    # Parse filename to get general info
    file_vars = get_file_variables(fn)
    if file_vars is None:
        print(f"Unexpected filename. Skipping file...")
        continue

    # Write general info to dictionary
    new_obj = {}
    new_obj["general"] = {}
    for key in file_vars:
        new_obj["general"][key] = file_vars[key]

    # Now get the rest of the general info fields from settings dataframe
    overview_df = overview_measures[overview_measures["short_name"] == file_vars["short_name"]]
    if len(overview_df)==0:
        print(f"Problematic measure: short_name {file_vars['short_name']} not found in xlsx file")
        continue
    overview_info = overview_df.iloc[0]

    # Then write the rest of the general info fields to the dictionary
    overview_dict = {}
    fields_to_add = ["long_name", "description", "instruction", "reference", "doi", "measure_category"
    for key in fields_to_add:
        overview_dict[key] = overview_info[key]
    new_obj["general"].update(overview_dict)
```

```
# ---
# 2. VARIABLES AND LEVELS
# ---

# Read info from .sav file using pyreadstat (instead of Pandas, since it provides access to sav file metadata)
df, meta = prs.read_sav(fn, encoding="latin1")

# Get column names (i.e. variables) as list
cols = list(df.columns)
# Get column labels (i.e. descriptions), variable options, and variable option keys
labels = meta.column_names_to_labels
variable_options = meta.variable_value_labels
variable_option_keys = variable_options.keys()

# New object variables field = empty dictionary
new_obj["variables"] = {}
# Loop through column names (i.e. variables), skipping already written fields
cols_to_skip = ["Cohort", "Wave"]
for colname in cols:
    if colname in cols_to_skip:
        continue

    new_obj["variables"][colname] = {}
    new_obj["variables"][colname]["label"] = labels[colname]

    if colname in variable_option_keys:
        new_obj["variables"][colname]["levels"] = variable_options[colname]
    else:
        new_obj["variables"][colname]["levels"] = "n/a"
# New json filename
json_fn = fn.replace('sav', 'json')
# Write dictionary to file
with open(json_fn, 'w') as f:
    json.dump(new_obj, f)
```

Tips

Be kind to your future self:
Manage data before you have data

- **Data manager from the start**
- **Make JSON files & measure overview from the start**
- **Use our format as a starting point**
  templates available on the website



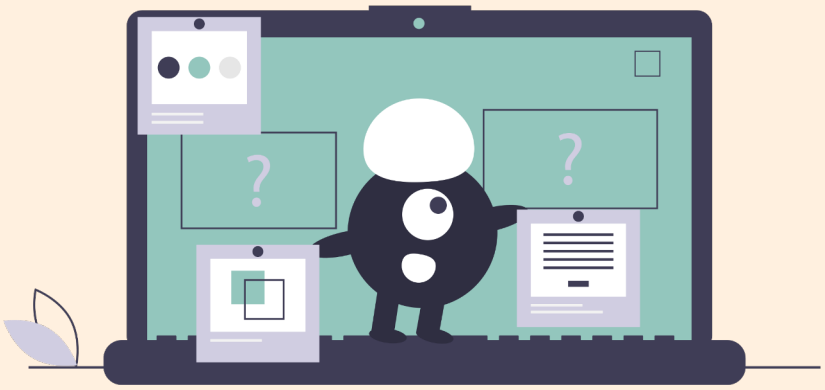
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Challenge II: How to get from raw data to structured data?





Challenges

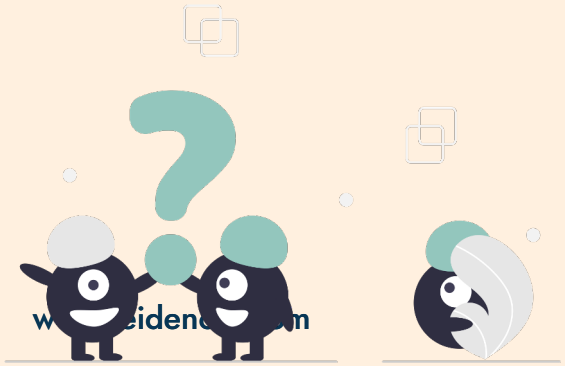


1. Data format conversion
2. Missing data
3. Version control?

Struggling with data

Start early!

Find a method… or make one!



BIDS


Remi Gau
@RemiGau
Follow
This needed to be done.
#BIDS is the #mariekondo for #fMRI #MRI #EEG #MEG #DataCuration.


This one does not spark joy.
dicomdir/
1208200617178_22/
1208200617178_22_8973.dcm
1208200617178_22_8943.dcm
1208200617178_22_2973.dcm
1208200617178_22_8923.dcm
1208200617178_22_4473.dcm
1208200617178_22_8783.dcm
1208200617178_22_7328.dcm
1208200617178_22_9264.dcm
1208200617178_22_9967.dcm
1208200617178_22_3894.dcm
1208200617178_22_3899.dcm
1208200617178_23/
1208200617178_24/
1208200617178_25/


This one sparks joy.
my_dataset/
participants.tsv
sub-01/
anat/
sub-01_T1w.nii.gz
func/
sub-01_task-rest_bold.nii.gz
sub-01_task-rest_bold.json
dwi/
sub-01_dwi.nii.gz
sub-01_dwi.json
sub-01_dwi.bval
sub-01_dwi.bvec
sub-02/
sub-03/
sub-04/

BIDS - why do we use it?

Tidy

Easy collaboration



Automation

Future proof

Challenges - data format conversion

- Which converter to use?

- Bidsify
  - Not supported anymore (changes)
  - Uses dcm2niix

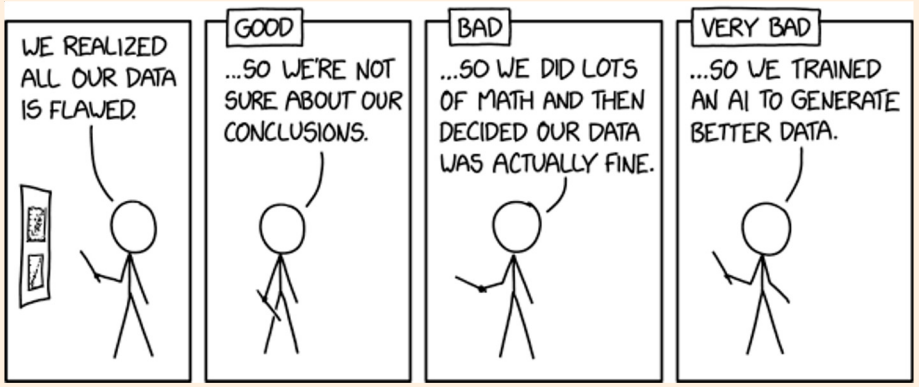
https://xkcd.com/2116/

Challenges - missing data

- Delays!
  - Can be mitigated

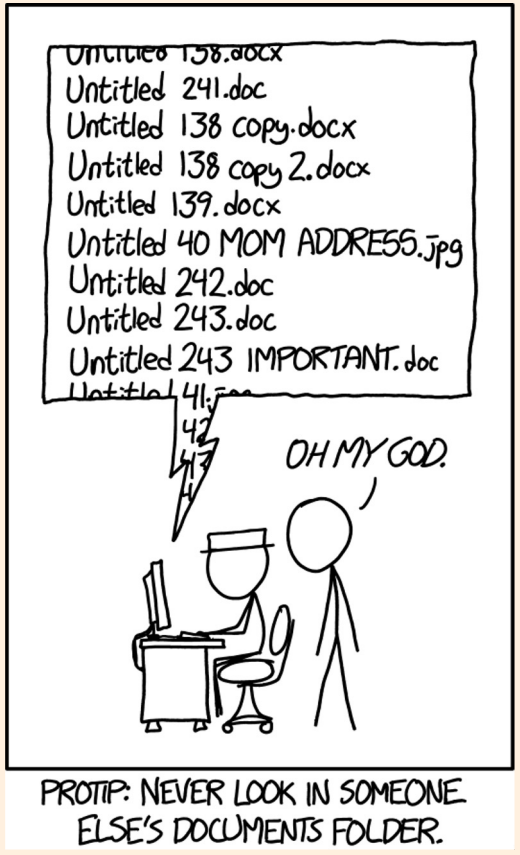
- Tagging files
  - Time to document and tag
  - Inject workflow logic: what to do?

    E.g. **ignore_missing=True; convert_existing_data=False; convert_t1w_only=True**



https://xkcd.com/2494/

Challenges - version control + provenance



Raw-to-structured

- ~100% scripted conversion
  - Ever changing
  - *git*, hosted on *GitHub*
- Source data
  - No version control
  - Rerun improved scripts
- Structured data
  - *DataLad*

https://xkcd.com/1459/

Tips

- **Software dependencies and workflow steps**
  - Upfront!
- **Community supported tools**
  - open source software ⇒ maintenance!
- **Lab notebook**
  - specifies data irregularities + reasons
- **Workflow logic**
  - automatic decision making (based on data irregularities)
- **Code AND data version control from the start**
  - raw source data, structured, processed
  - community supported tools

Challenge III: How to be open and protective?

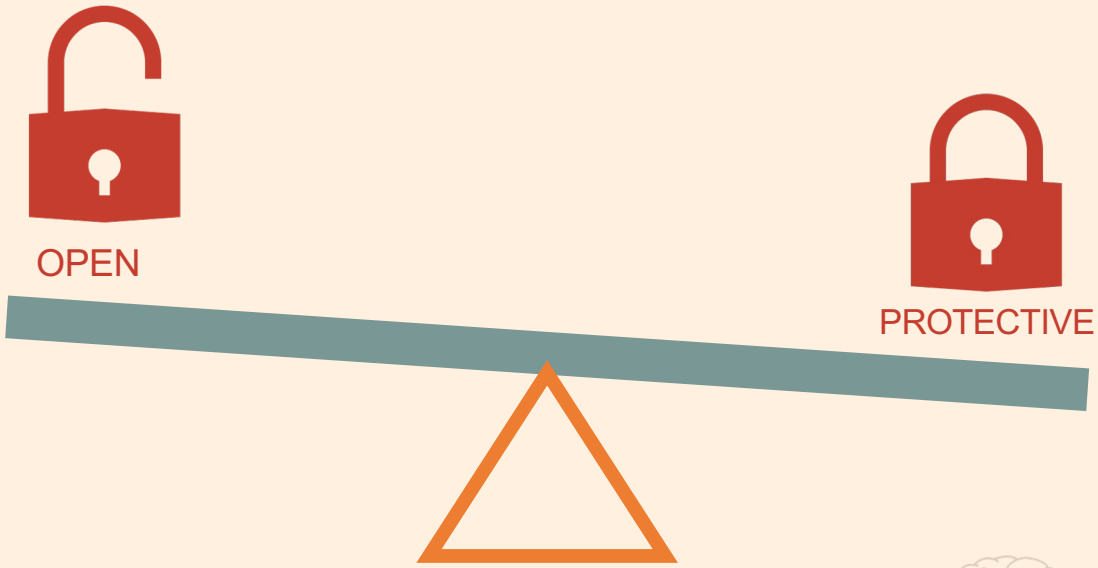




How to be open and protective?

The challenge:

GDPR was not in place when study started (9 years ago)



Findable and accessible... yet secure?

Open... yet protective?



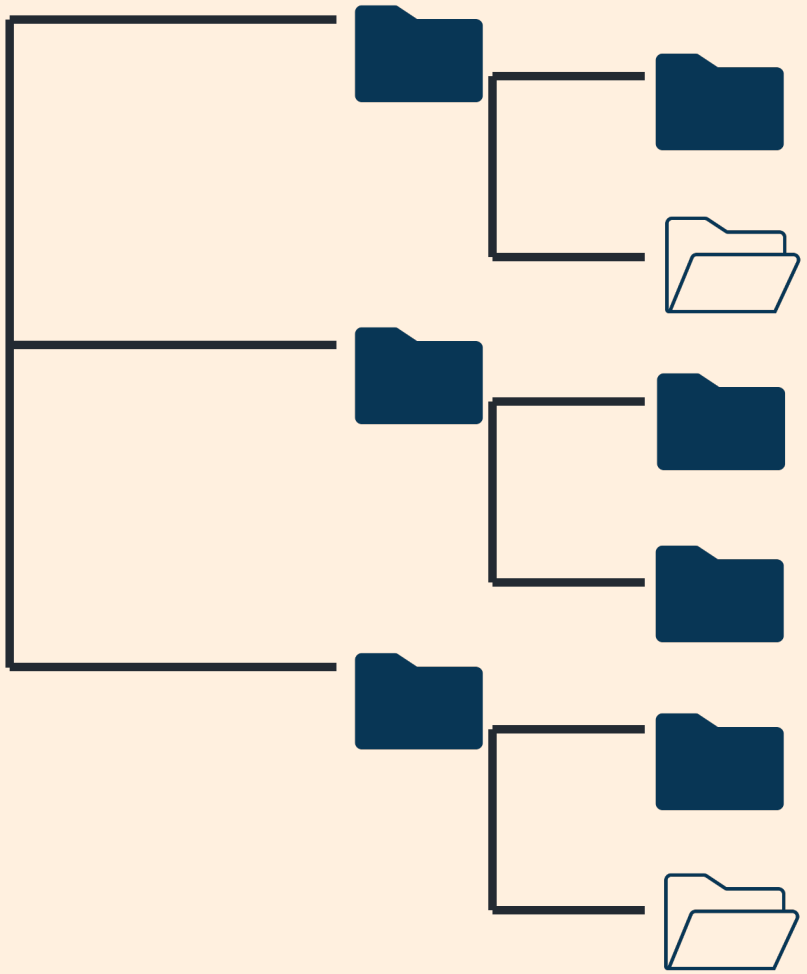
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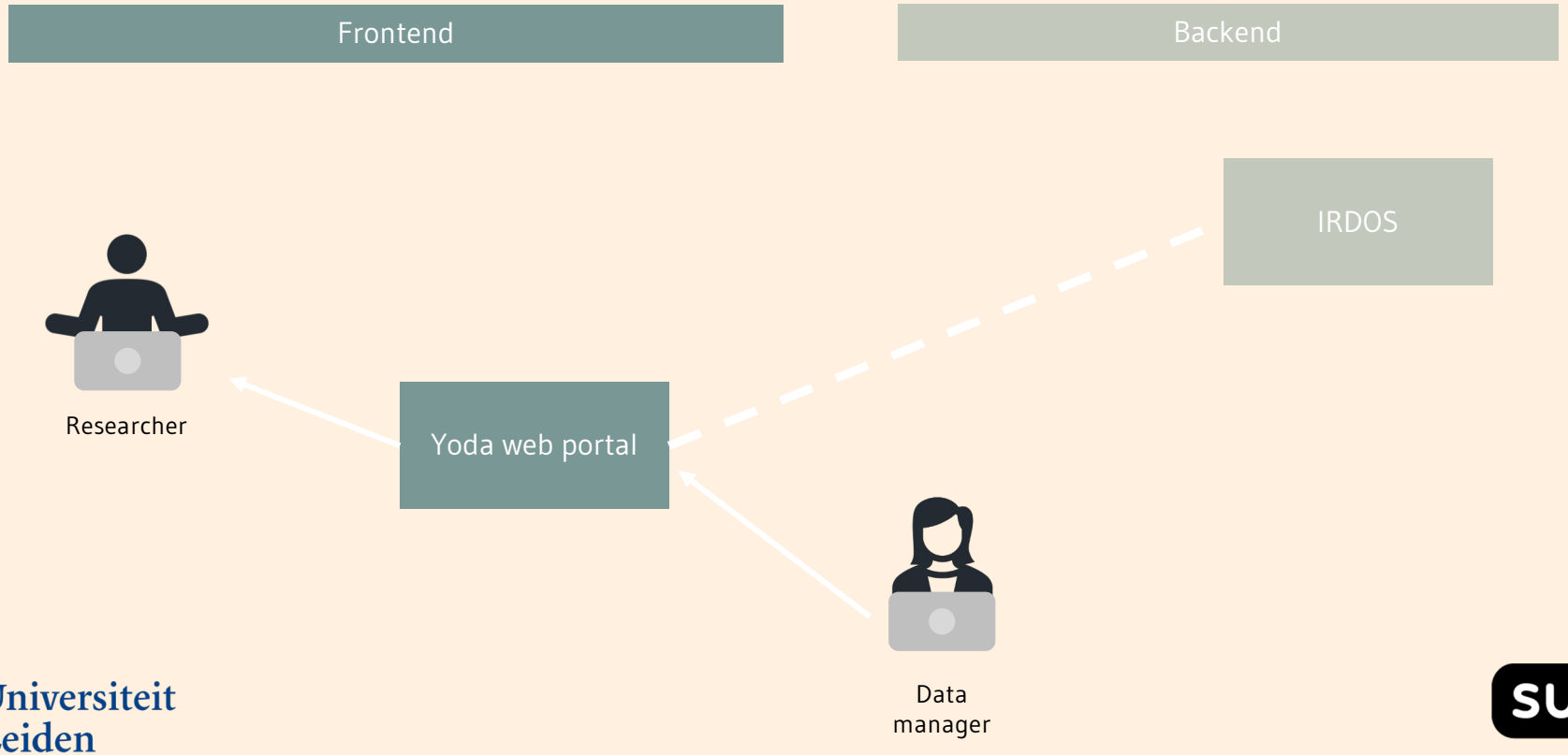
Why YODA (Data sharing platform)?

- You can be sure that your valuable data is **stored securely**;
- You **work** with your data **and archive it in one place**;
- You can **efficiently collaborate** with partners by giving them access to your data;
- Yoda is also suitable for **(privacy) sensitive data**;
- You can easily retrieve **previous versions** of your files if needed;
- You make your data **findable** for yourself and others by adding **metadata**;
- You can **publish** your data to make your work **visible** for others;
- You manage and share your data in c**ompliance to privacy rules** and data management regulations;
- You **contribute to Open Science**, because Yoda helps you to make your data findable for a broad audience;
- You can **work with your data from multiple locations** through the website portal;
- You help to **increase the (re)use of data** since Yoda makes it possible for you to make your own data accessible for others. You can also retrieve existing data to make use of it;
- You can use Yoda for **all kinds of digital source files**, like text files, videos, photos, excel files, R files, STATA, files, datasheets, geographical data, etc.

Restricted data sharing



Coming soon!

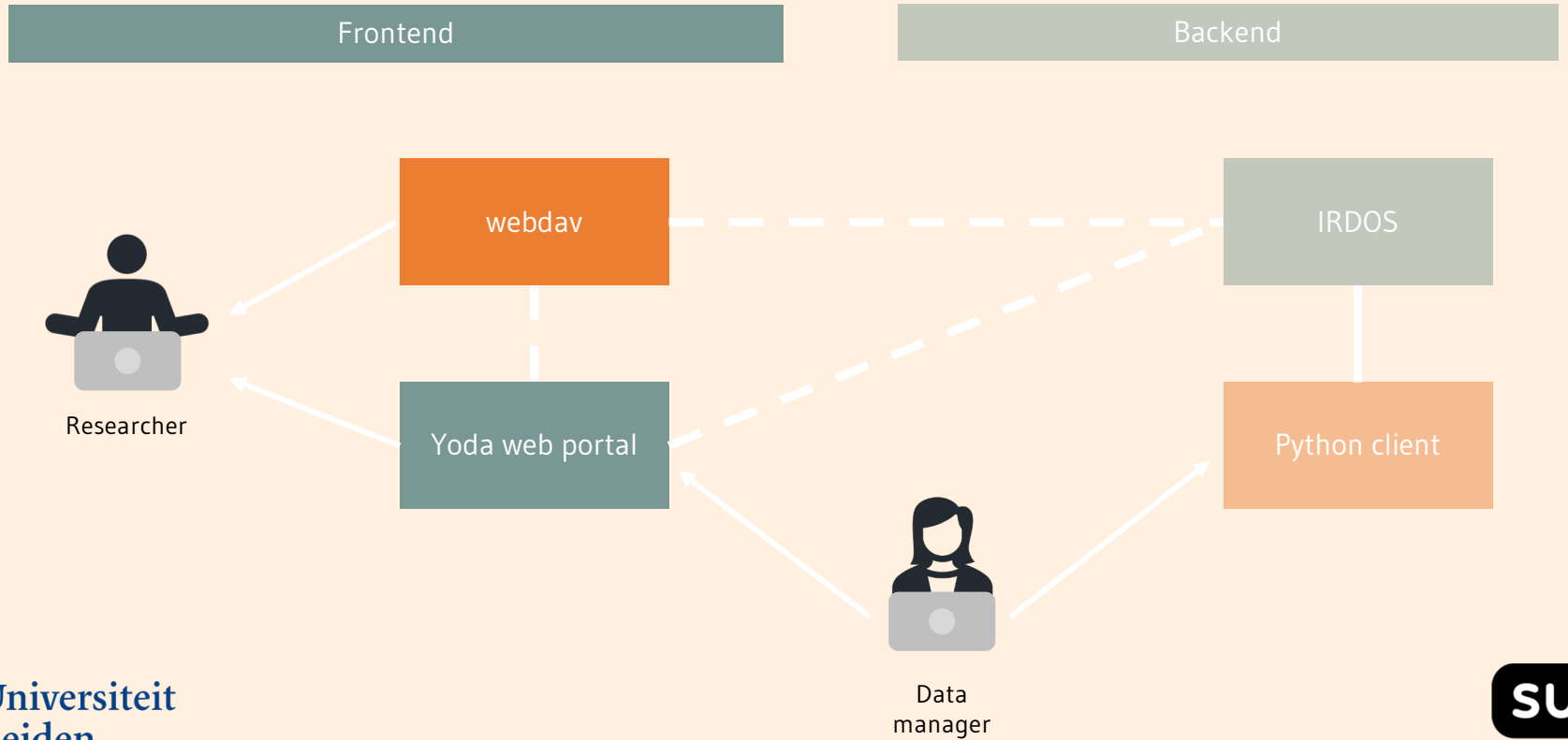


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SURF

Coming soon!

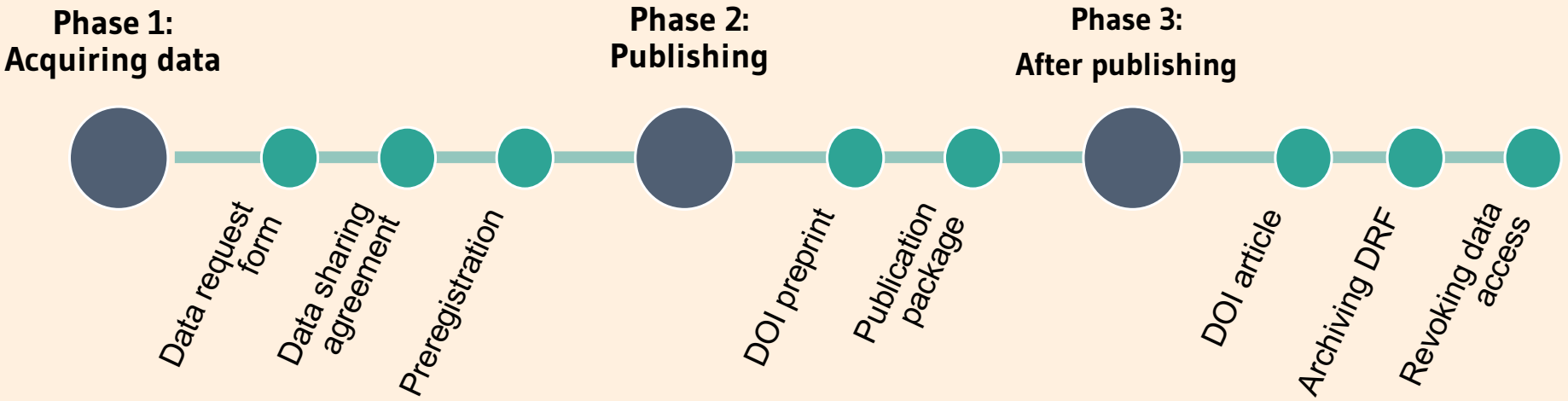


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SURF

Protocol for data sharing

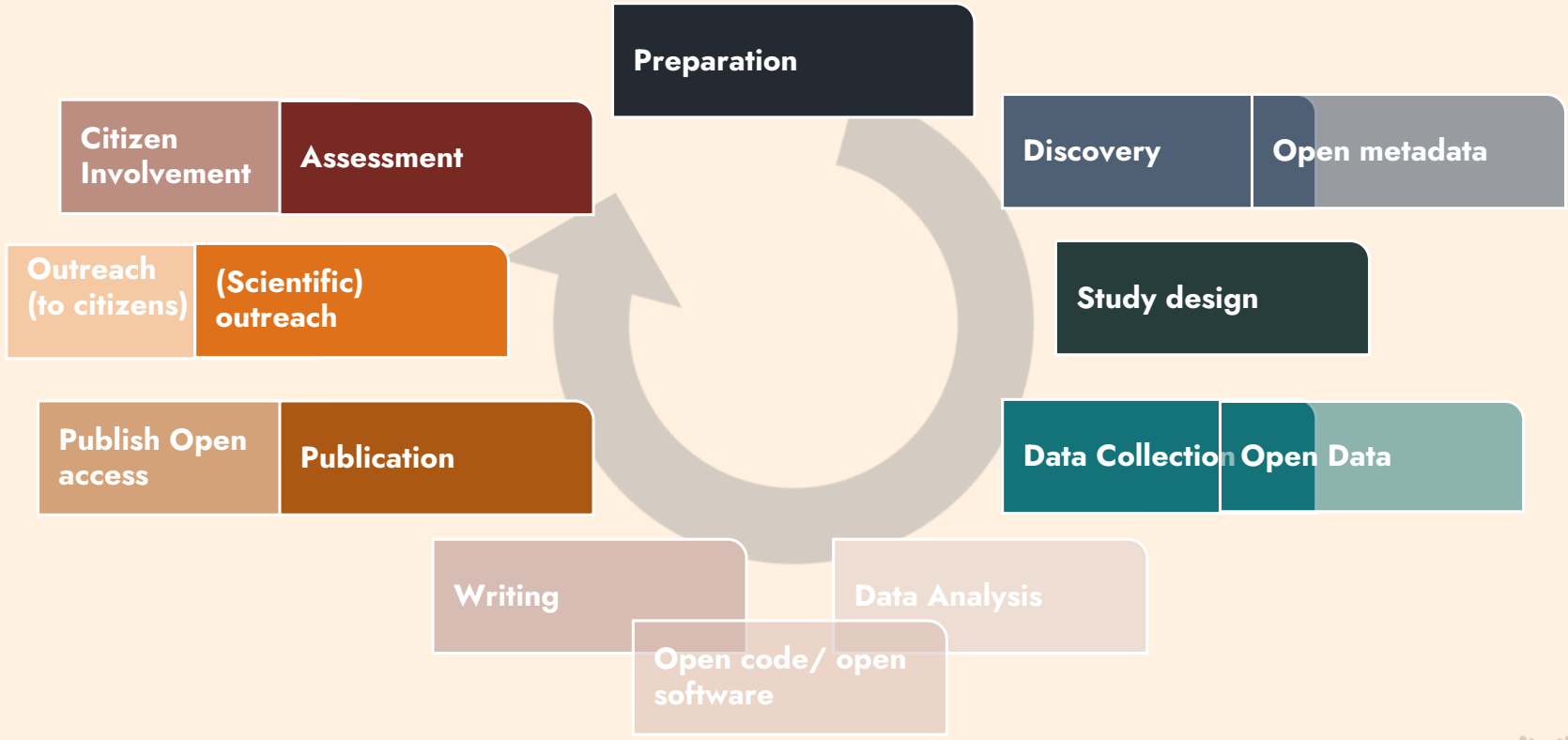


Challenge IV: How to be findable?

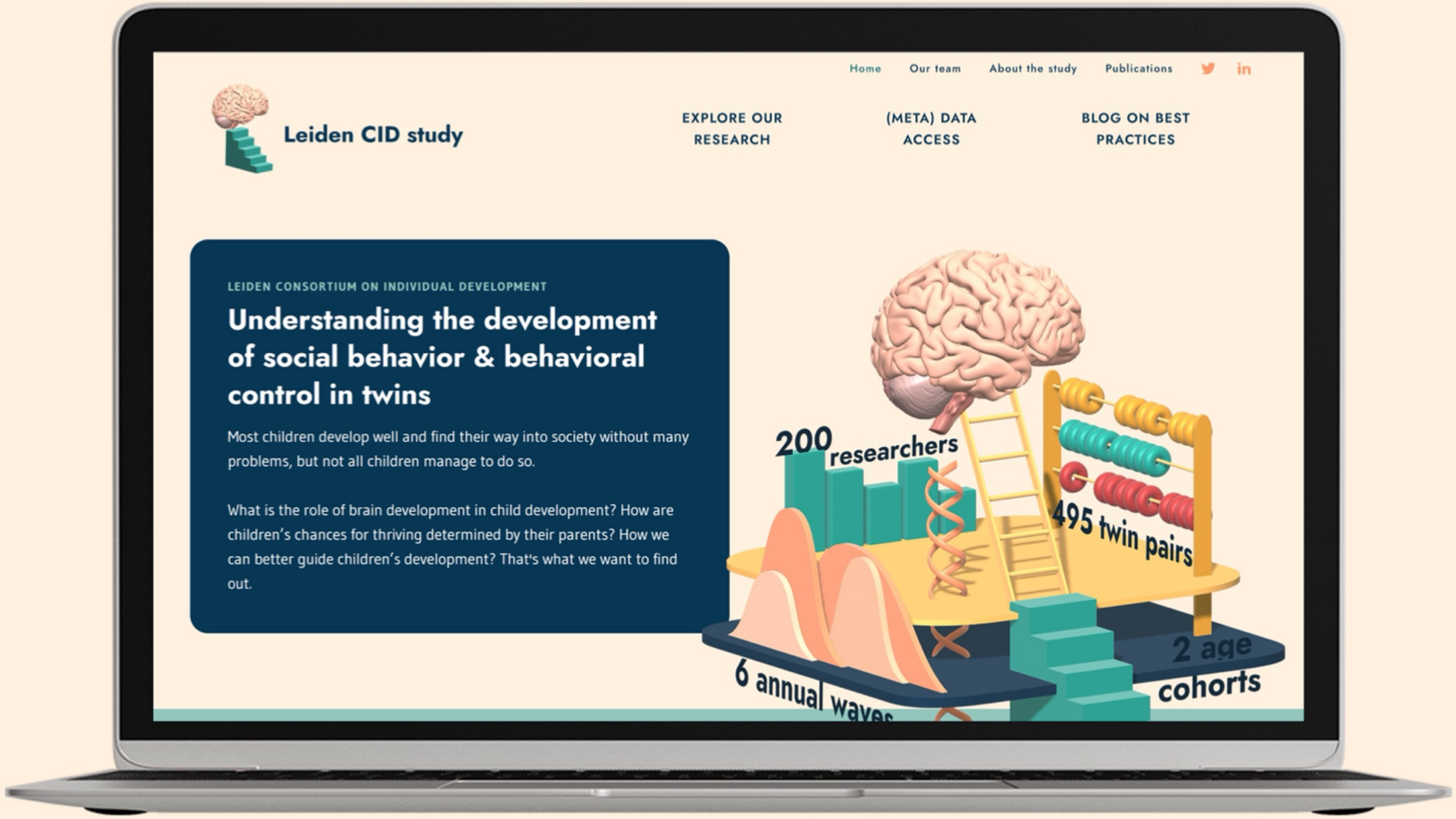




How to be findable and accessible?



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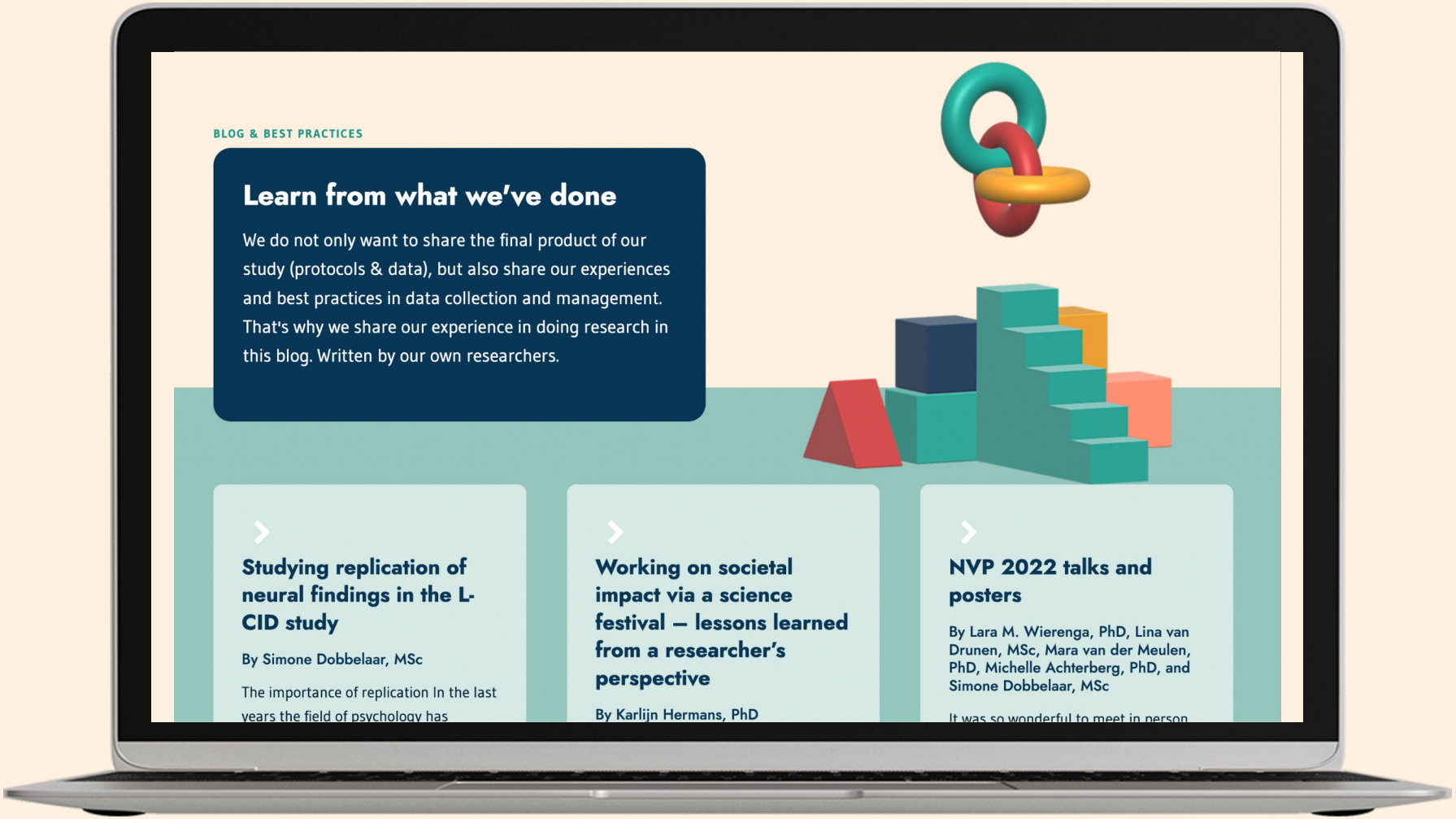
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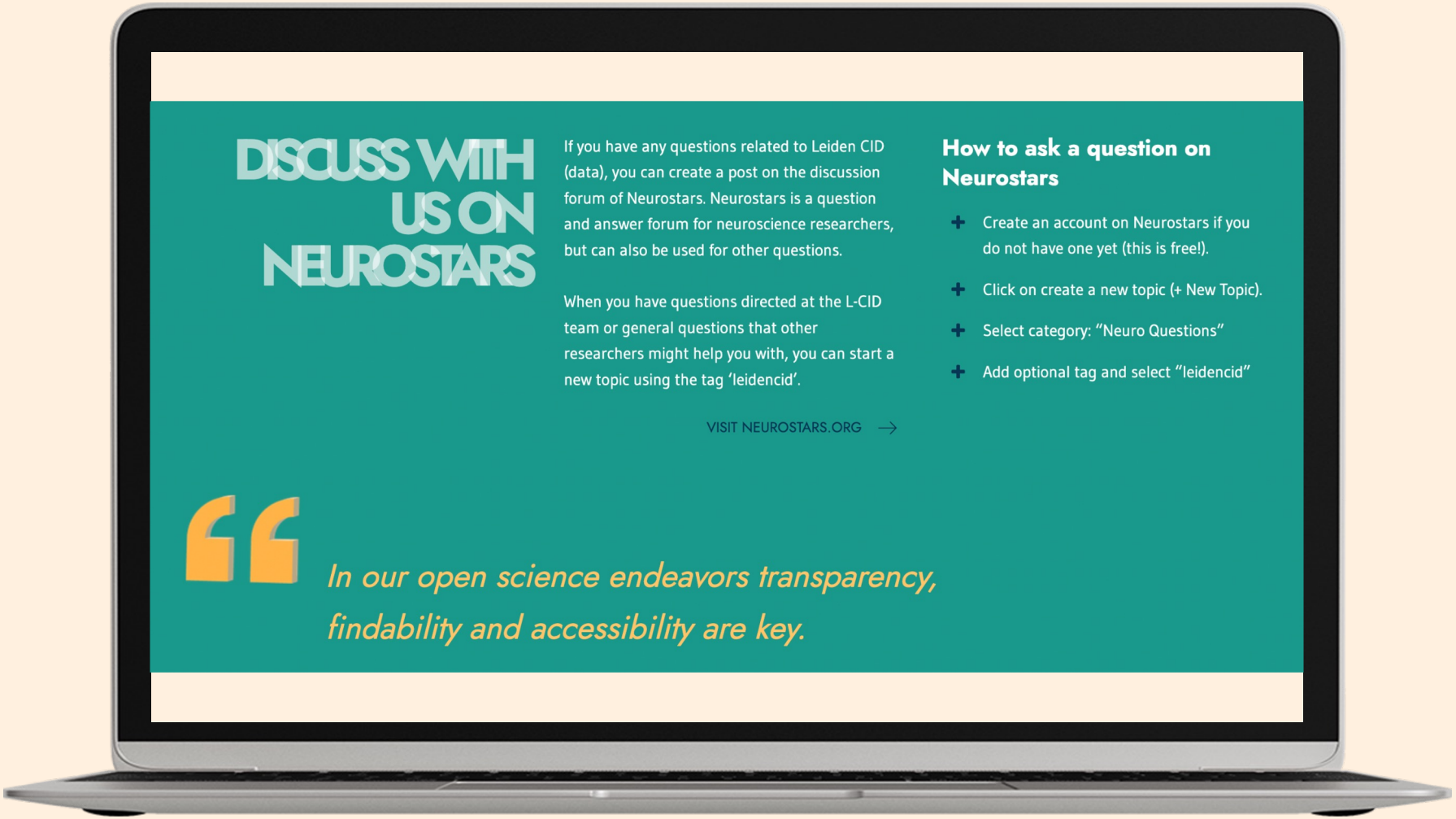
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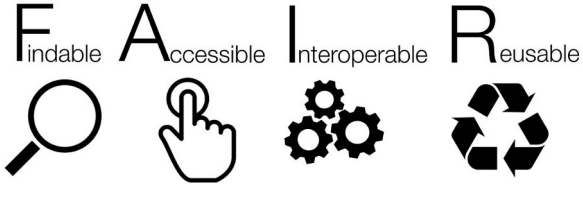
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DISCUSS WITH US ON NEUROSTARS
If you have any questions related to Leiden CID (data), you can create a post on the discussion forum of Neurostars. Neurostars is a question and answer forum for neuroscience researchers, but can also be used for other questions.
When you have questions directed at the L-CID team or general questions that other researchers might help you with, you can start a new topic using the tag 'leidencid'.
VISIT NEUROSTARS.ORG
How to ask a question on Neurostars
+ Create an account on Neurostars if you do not have one yet (this is free!).
+ Click on create a new topic (+ New Topic).
+ Select category: "Neuro Questions"
+ Add optional tag and select "leidencid"
In our open science endeavors transparency, findability and accessibility are key.

Let's talk metadata: Findability



Connecting Data in Child Development (CD2)



Findable CID data

Metadata standards

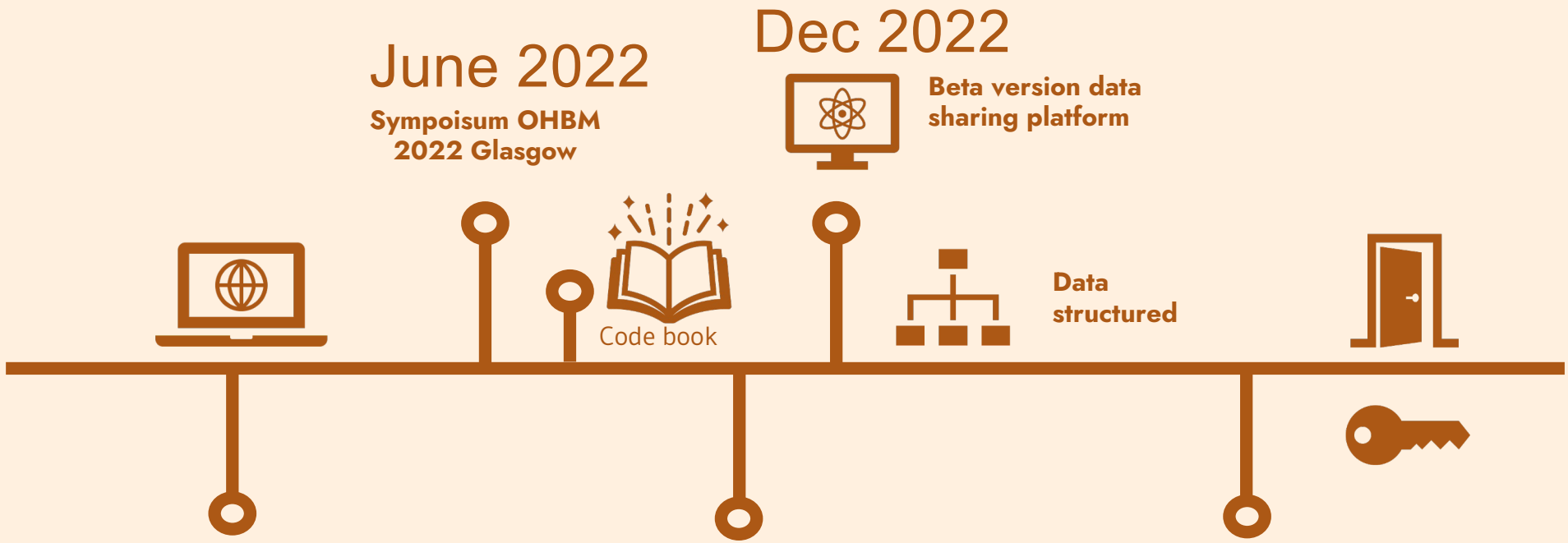
- CID metadata in searchable web portal

Consortium on Individual Development

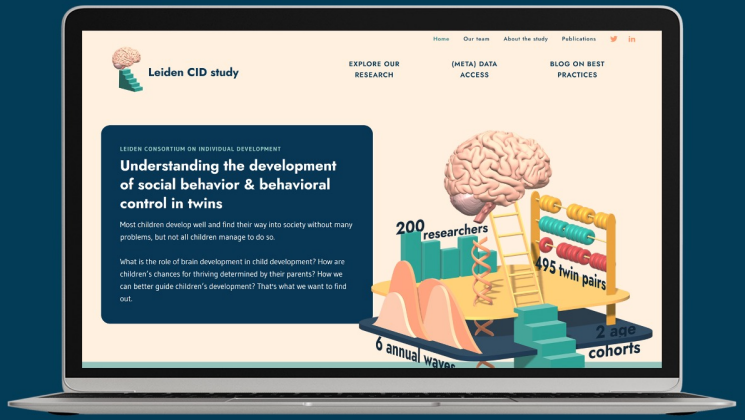
PDI-SSH

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OS roadmap



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